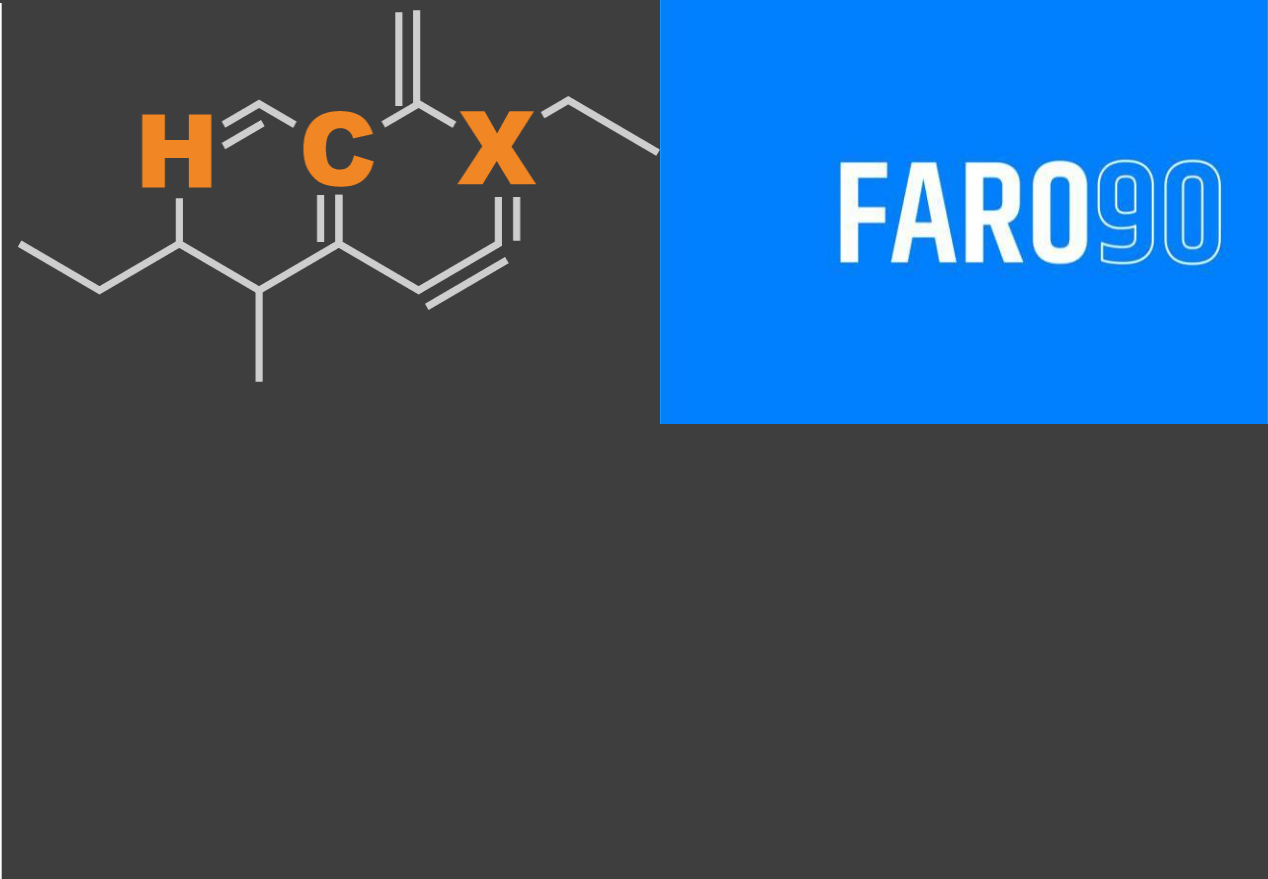




Ethanol Blending in Gasoline – Dominican Republic

Agosto, 2025

Ethanol Blending in Gasoline – Dominican Republic

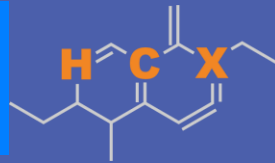
August 2025



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Ethanol Blending in Latin America

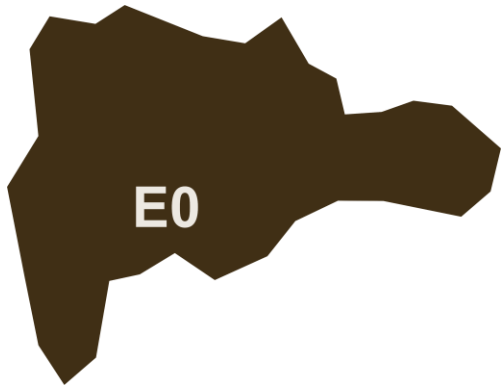
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



Ethanol Blending in Gasoline – Dominican Republic

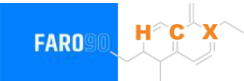


National demand is covered up by local production representing 15%, and imports from United States, Trinidad and Tobago, Caribbean, among other countries. In 2024, gasoline consumption was 562 million gallons (2,126 million liters). Regular gasoline consumption was 35% of total demand and premium 65%. The Dominican Government applies subsidies to gasoline prices.

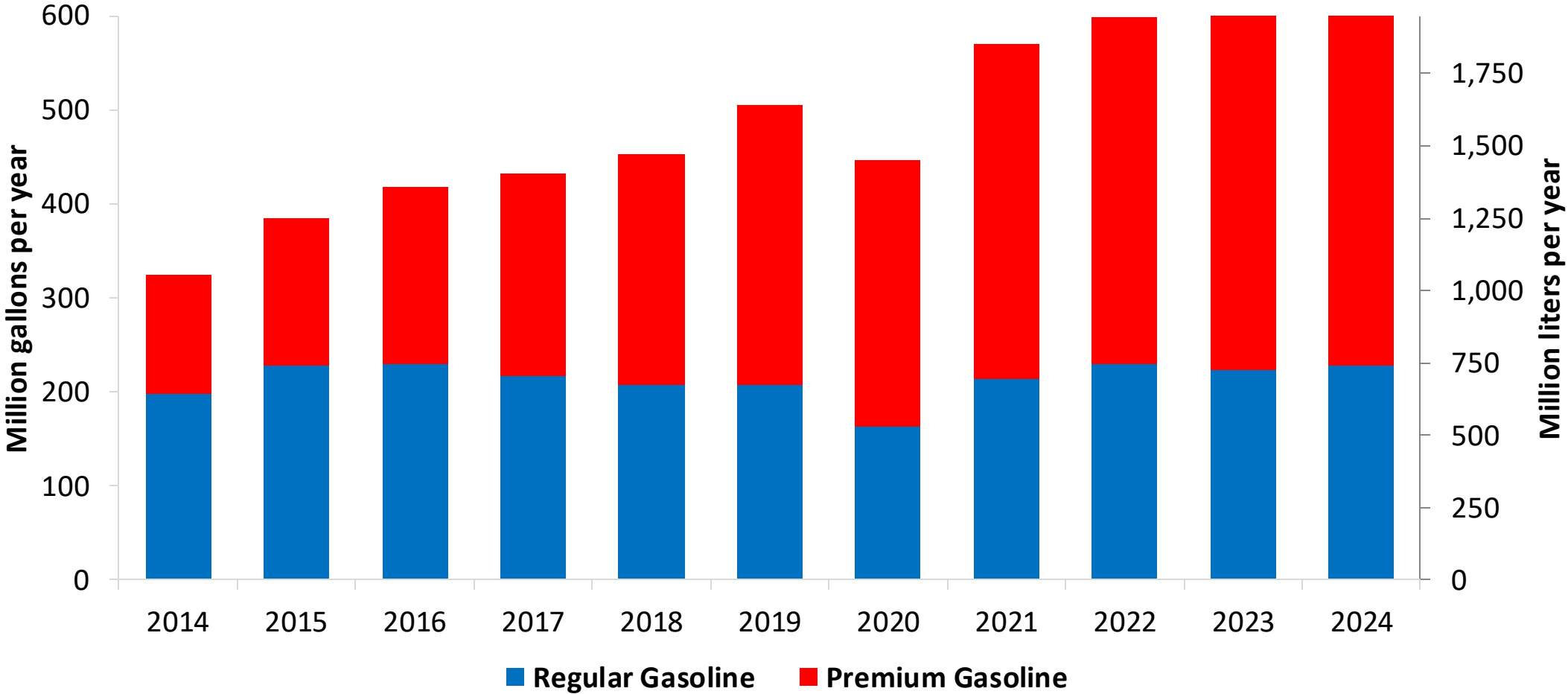
Dominican Republic does not currently have an ethanol mandate.

Source: DGII

Gasoline Demand in Dominican Republic

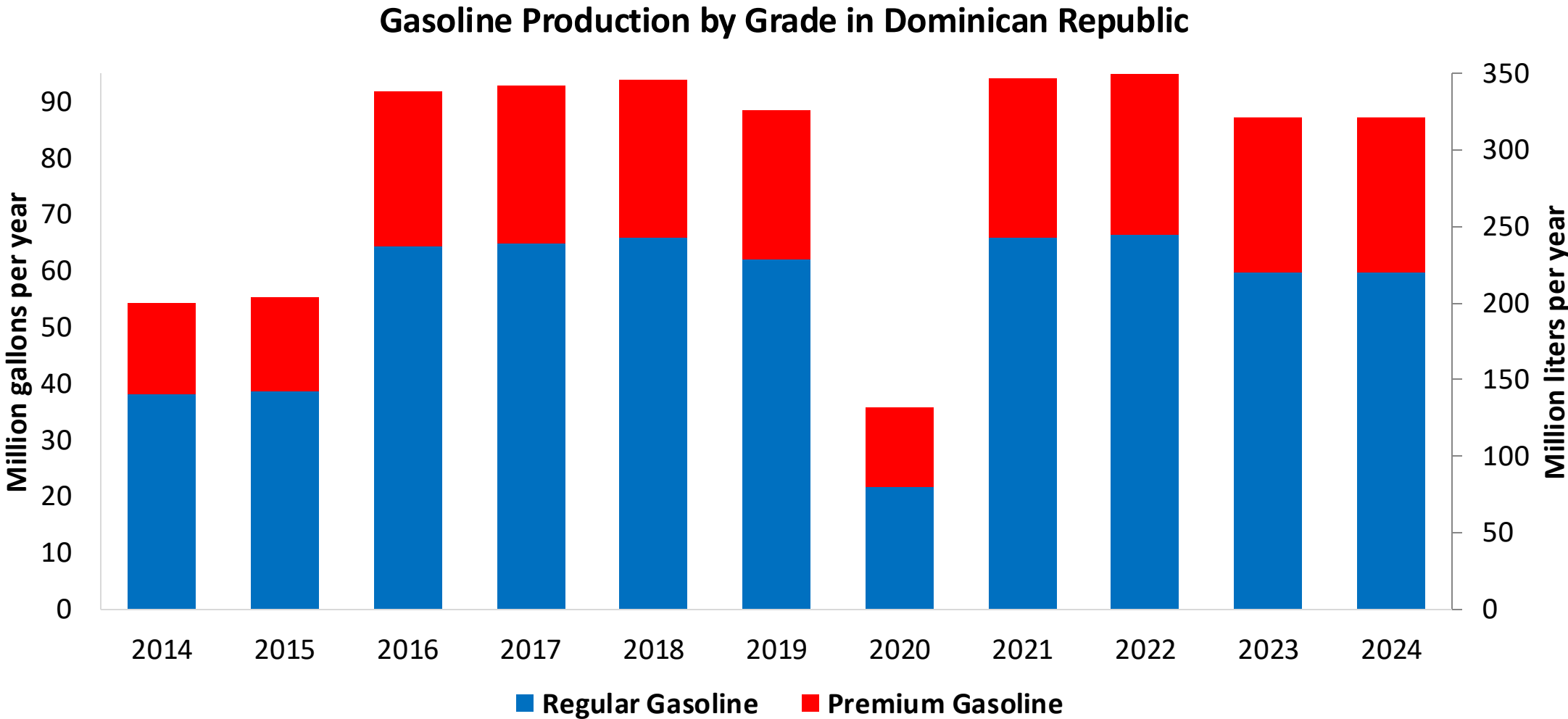


Gasoline Demand by Grade in Dominican Republic



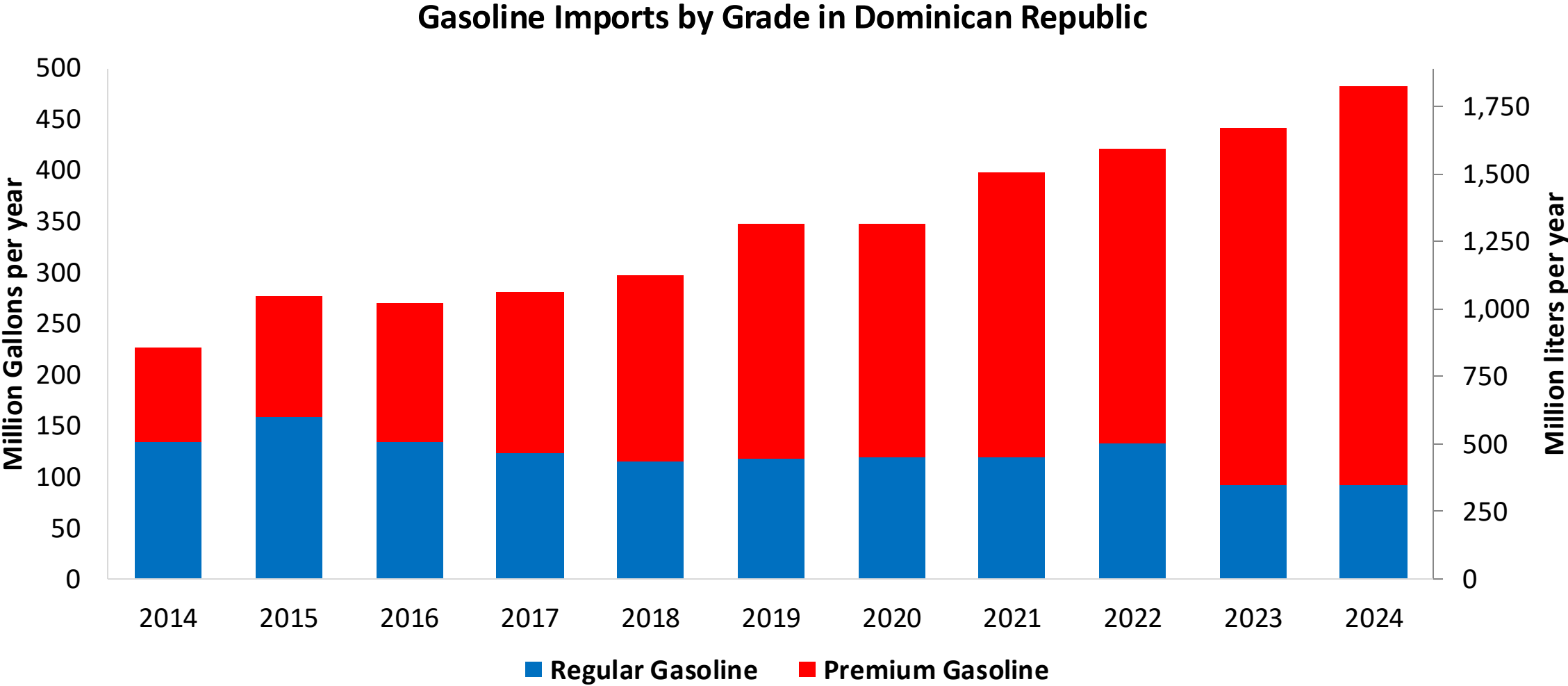
Source:
DGII

Gasoline Production in Dominican Republic



Source:
DGII

Gasoline Imports in Dominican Republic



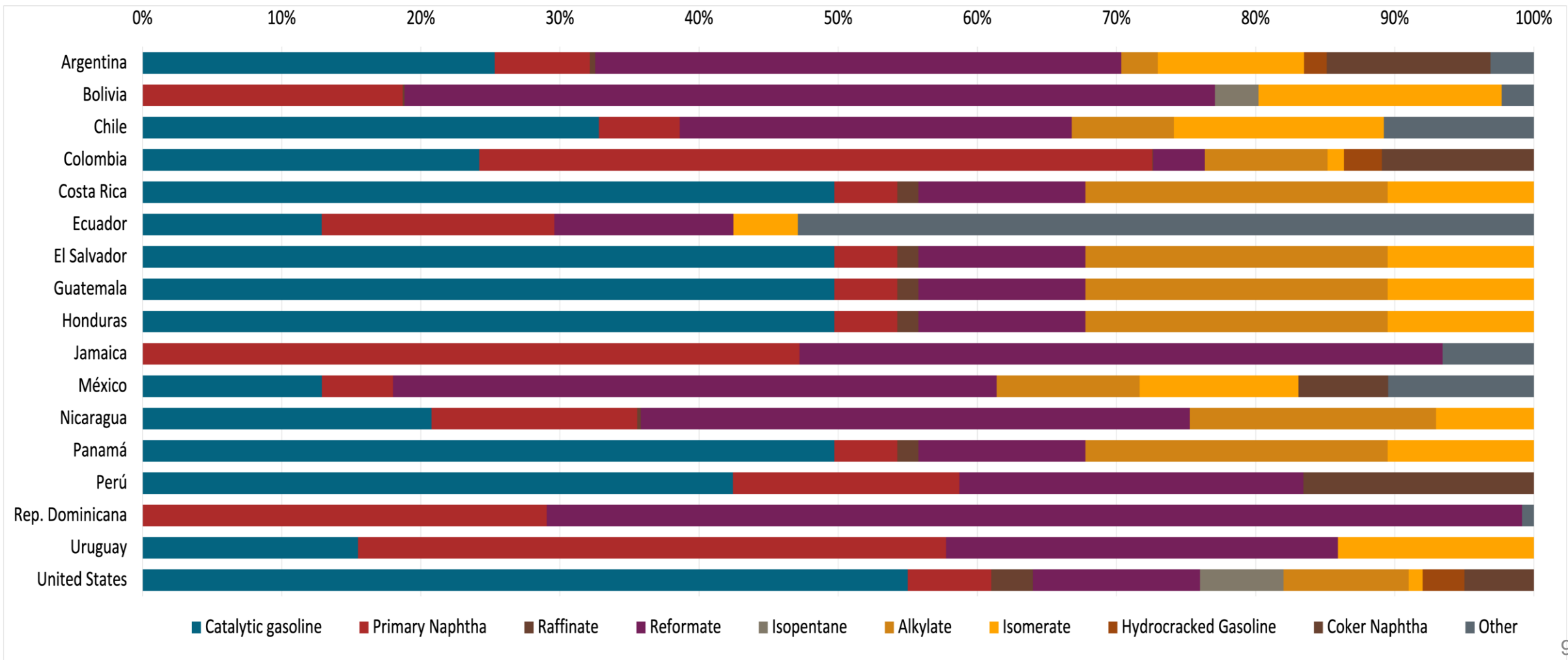
Source:
DGII

Gasoline Quality in Dominican Republic

Name	NORDOM 476 3rd Revision				EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2015				2017			
Applicability	Whole country	Whole country	Whole country	Whole country	All countries			
Selected Grade	Regular Gasoline	Premium Gasoline	Oxygenated Regular Gasoline	Oxygenated Premium Gasoline	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	-	-	-	-	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	-	-	-	-	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	-	-	-	-	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 8,3 mg/l	< 8,3 mg/l	< 8,3 mg/l	< 8,3 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 89	> 95	> 90	> 96	> 95	> 95	> 98	> 98
MON	> 76	> 82	> 77	> 83	> 85	> 88	> 85	> 88
AKI								
Sulfur Content	< 1.500 mg/kg	< 1.500 mg/kg	< 1.500 mg/kg	< 1.500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content			< 3.5 %m/m	< 3.5 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)			< 10 %v/v	< 10 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 61 kPa	< 61 kPa	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)								
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



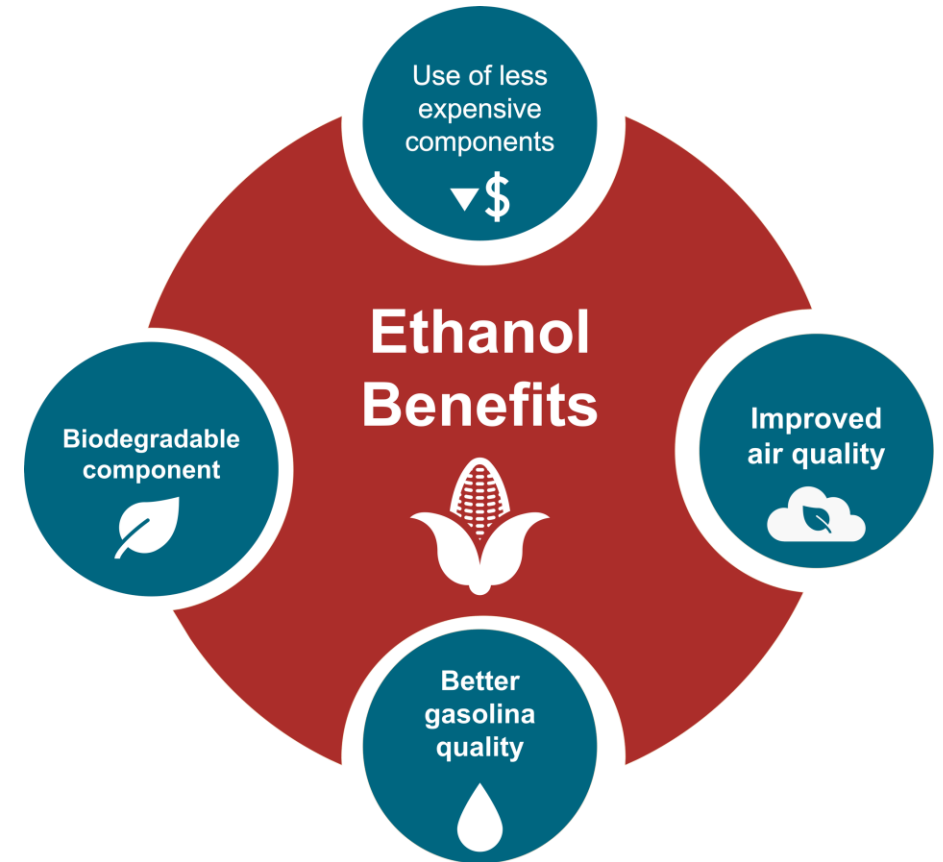
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

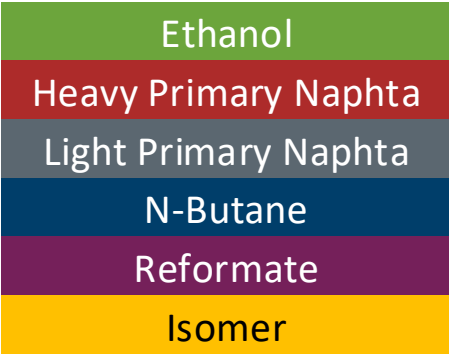
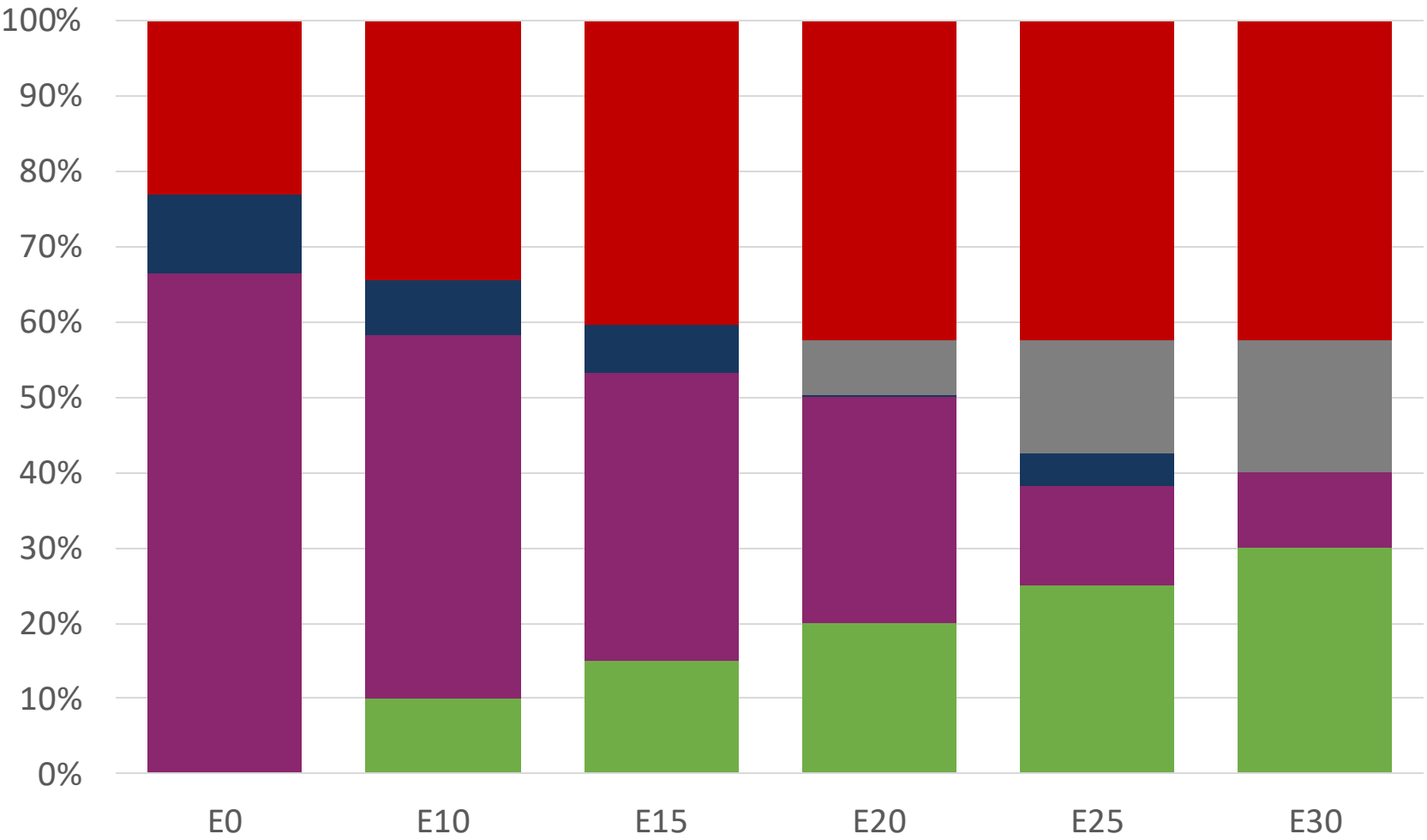
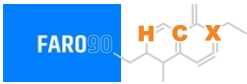
The blending model uses gasoline component spot average 2024 prices and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



The FARO90 logo is shown in a blue box, and a chemical structure is displayed next to it.



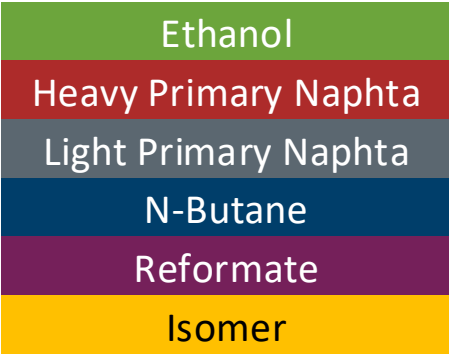
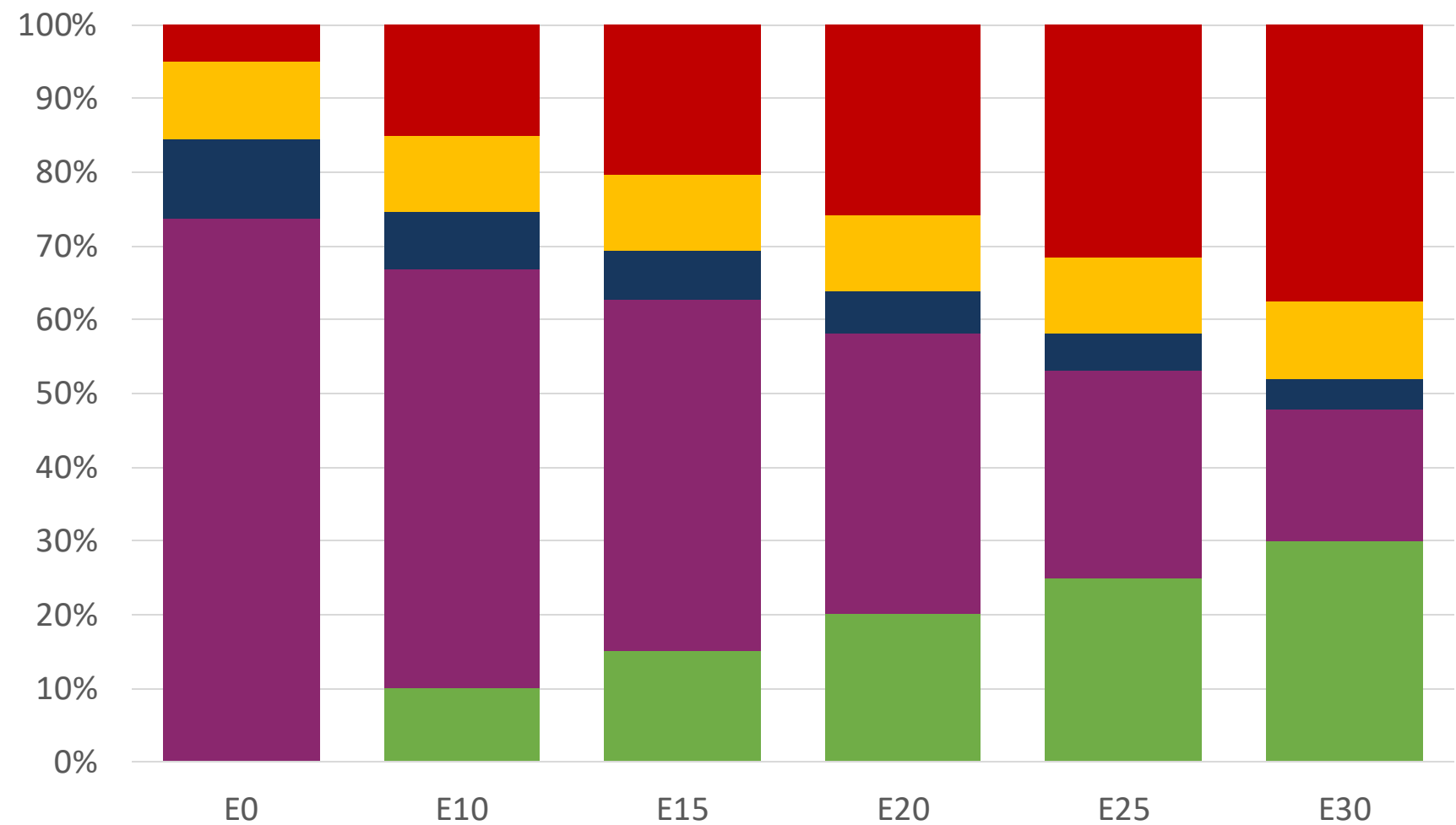
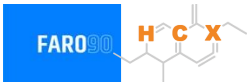
Dominican Republic – Regular – Constante Octane



Octane (RON)	90.0	90.0	90.0	90.0	90.0	91.8
Price (US\$/gal)	\$ 2.357	\$ 2.060	\$ 1.894	\$ 1.800	\$ 1.549	\$ 1.539

Source: Faro90

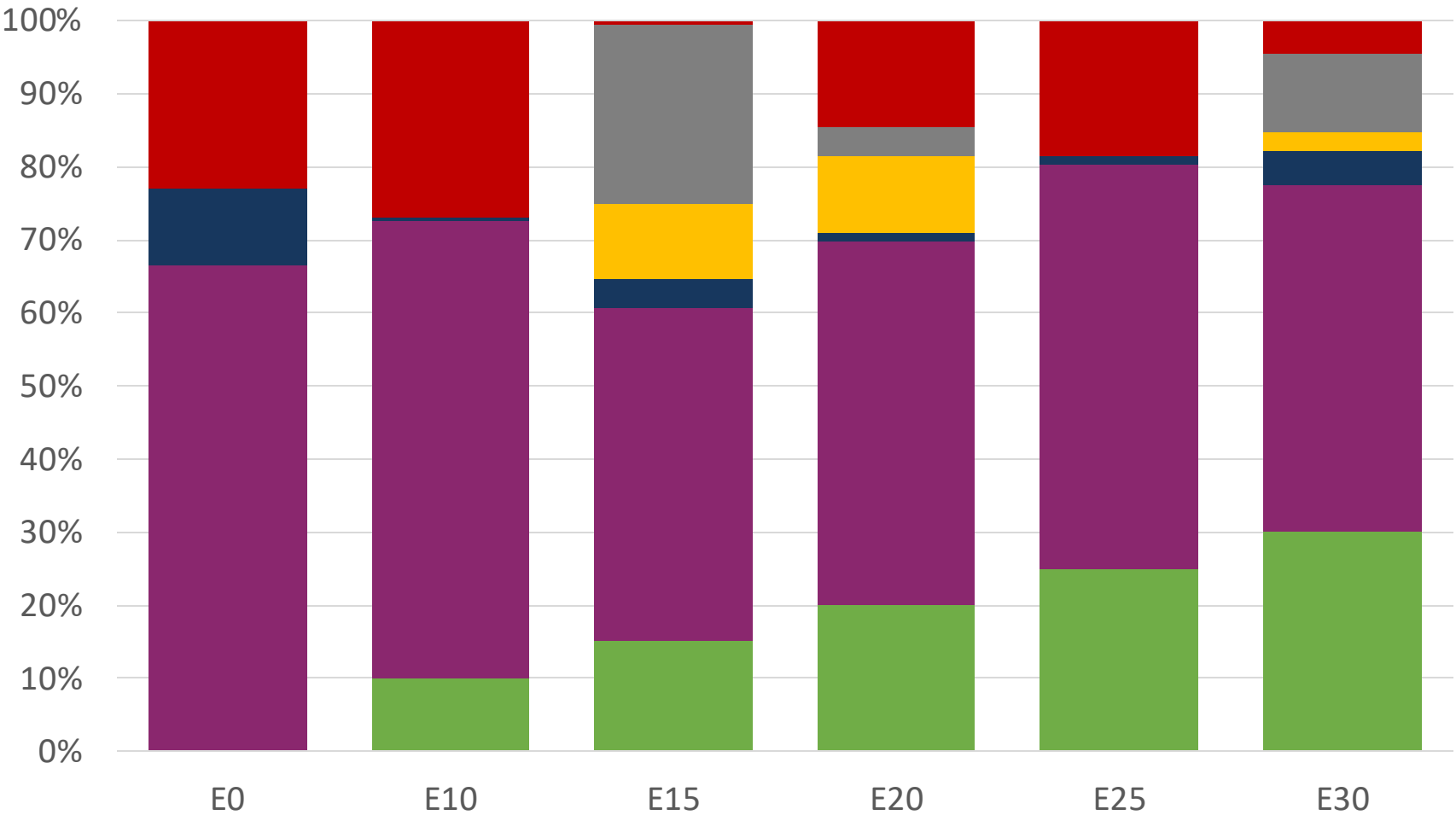
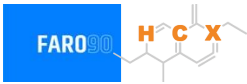
Dominican Republic– Premium – Constant Octane



Octane (RON)	96.0	96.0	96.0	96.0	96.0	96.0
Price (US\$/gal)	\$ 2.665	\$ 2.395	\$ 2.246	\$ 2.088	\$ 1.922	\$ 1.749

Source: Faro90

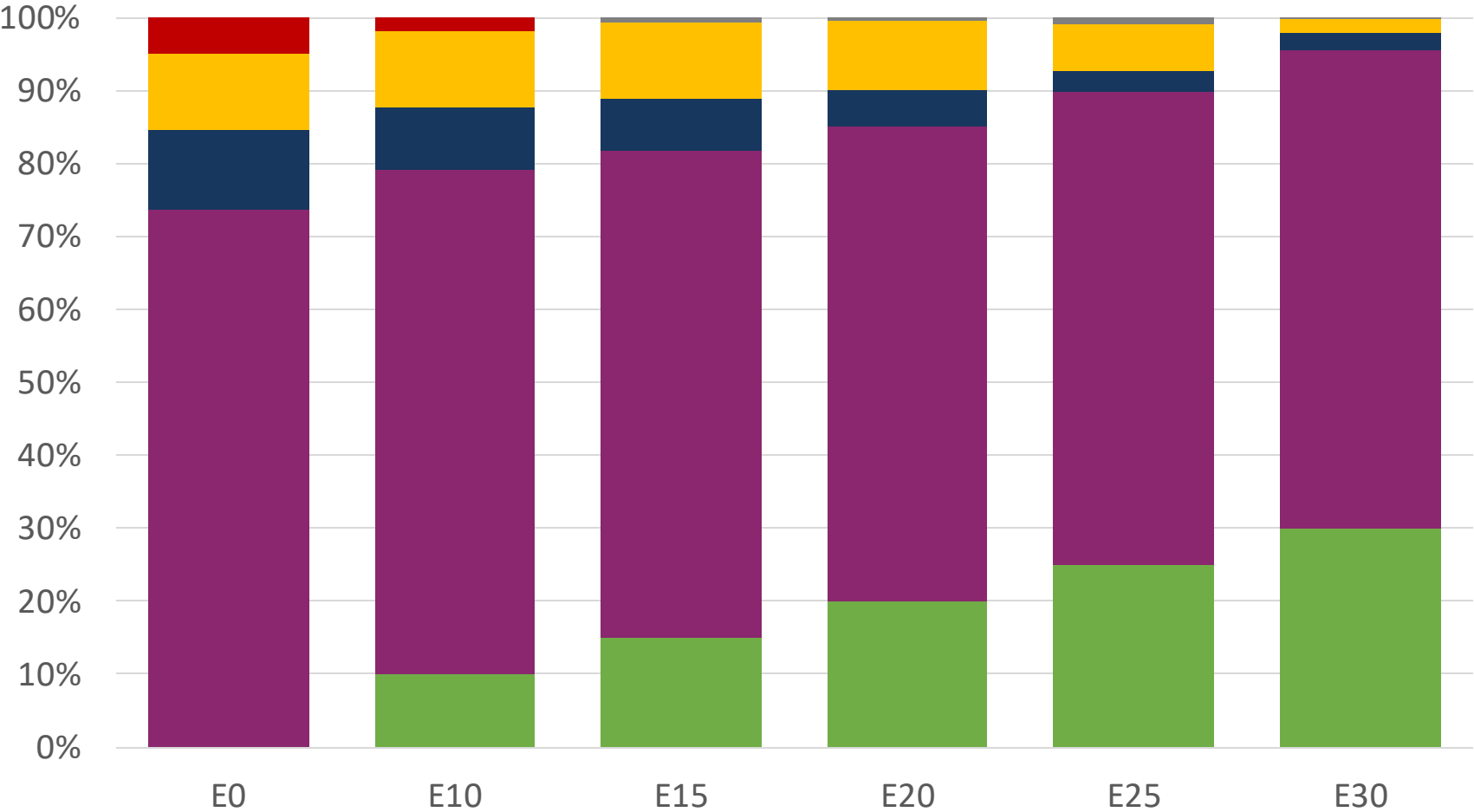
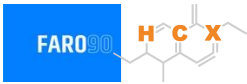
Dominican Republic – Regular – Increased Octane



Octane (RON)	90.0	93.2	96.8	99.0	101.3	104.3
Price (US\$/gal)	\$ 2.357	\$ 2.357	\$ 2.357	\$ 2.357	\$ 2.357	\$ 2.357

Source: Faro90

Dominican Republic – Premium – Increased Octane



Octane (RON)	96.0	100.8	102.9	104.6	106.3	108.2
Price (US\$/gal)	\$ 2.665	\$ 2.665	\$ 2.665	\$ 2.665	\$ 2.665	\$ 2.665

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

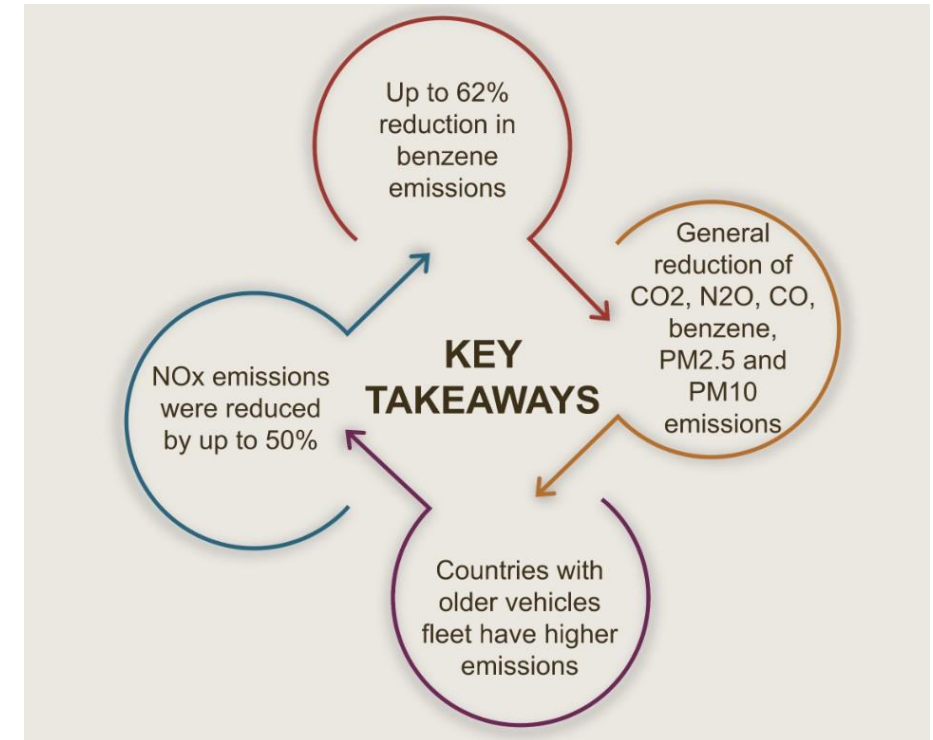
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

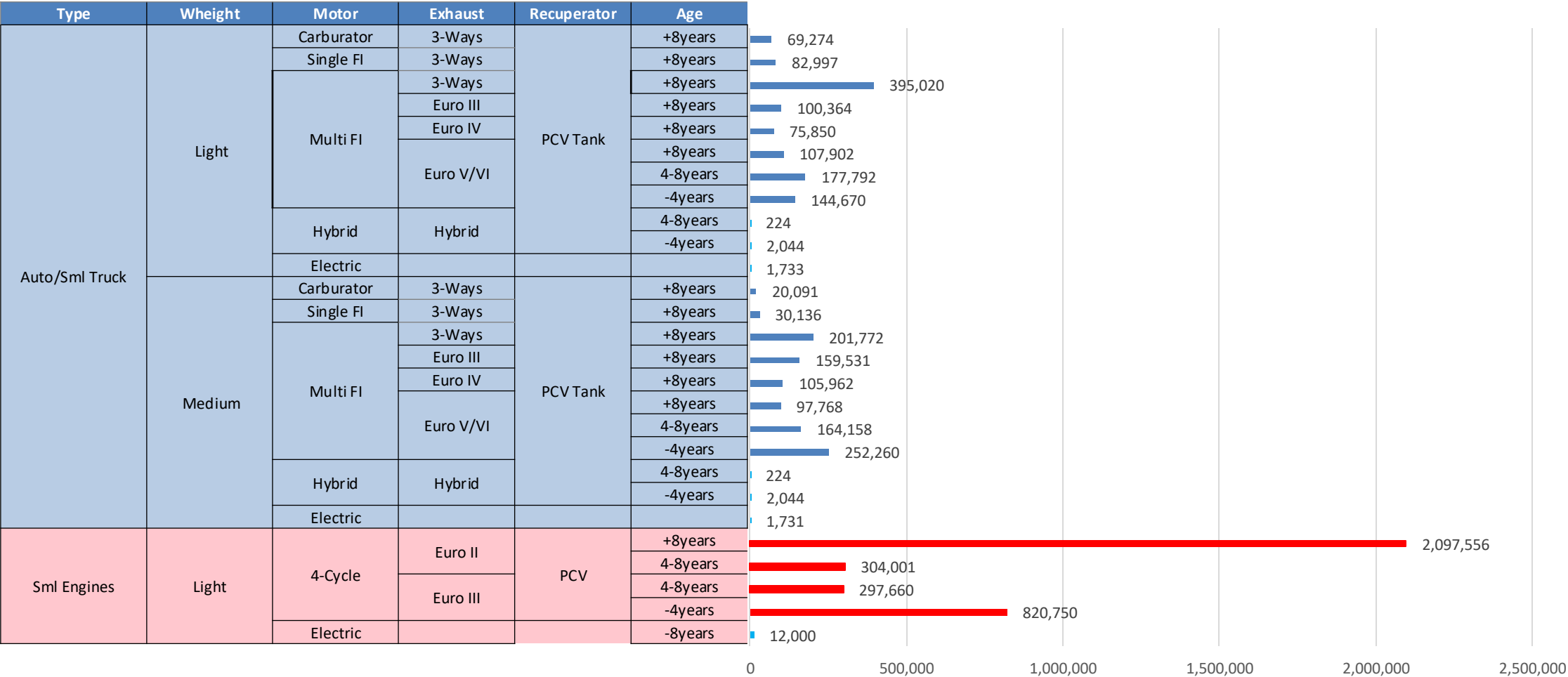
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicular Fleet in Dominican Republic

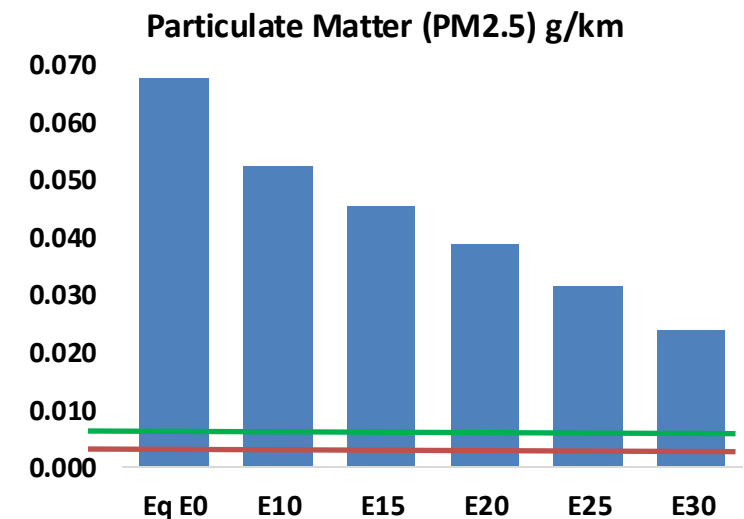
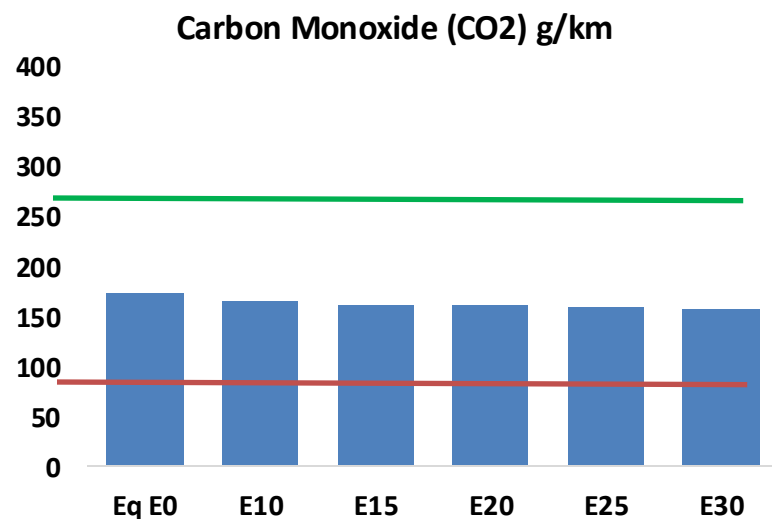
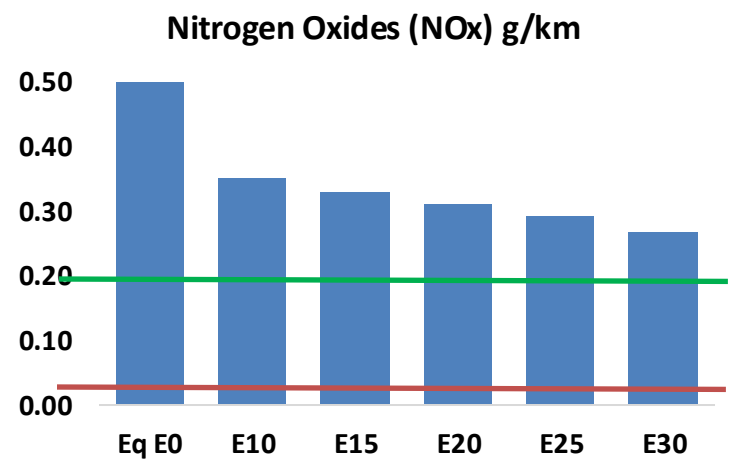


Gasoline Vehicular Fleet: 5,710,050

Average Age: 10 years

Motorcycles: 62%

United States



Dominican Republic – Vehicular Fleet



Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	725,082	651,131	577,181	513,029	446,835	395,443	341,040	-20%	-38%
CO2	16,771,425	16,340,160	15,932,854	15,612,520	15,454,006	15,384,904	15,101,316	-5%	-8%
VOC	108,194	100,531	92,869	86,717	80,585	75,888	69,557	-14%	-26%
NOx	48,234	36,176	33,764	31,823	30,011	28,014	25,902	-30%	-38%
SOx	368	340	313	289	266	242	216	-15%	-28%
NH3	6,342	6,280	6,217	6,246	6,317	6,366	6,407	-2%	0%
N2O	326	324	323	324	331	335	339	-1%	2%
CH4	27,917	27,917	27,917	28,336	28,950	29,481	29,983	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	697	655	612	579	547	523	490	-12%	-21%
Acetaldehyde	2,063	2,368	3,475	5,291	7,209	8,238	9,734	68%	249%
Formaldehyde	8,035	7,812	9,107	10,693	11,203	12,393	13,492	13%	39%
Benzene	2,126	2,043	1,936	1,905	1,889	1,807	1,748	-9%	-11%
PM2.5	1,222	1,085	948	824	699	567	430	-22%	-43%
PM10	5,261	4,671	4,082	3,545	3,010	2,443	1,851	-22%	-43%

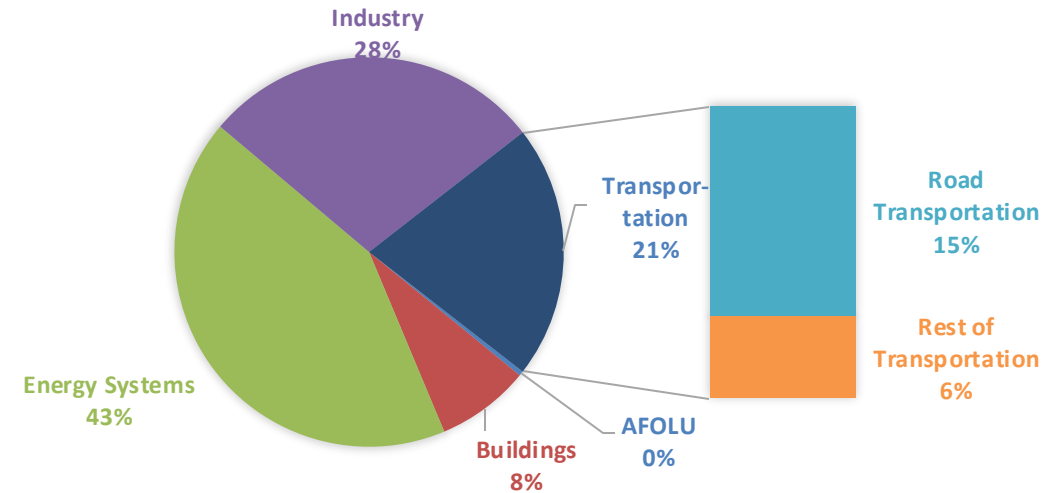
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

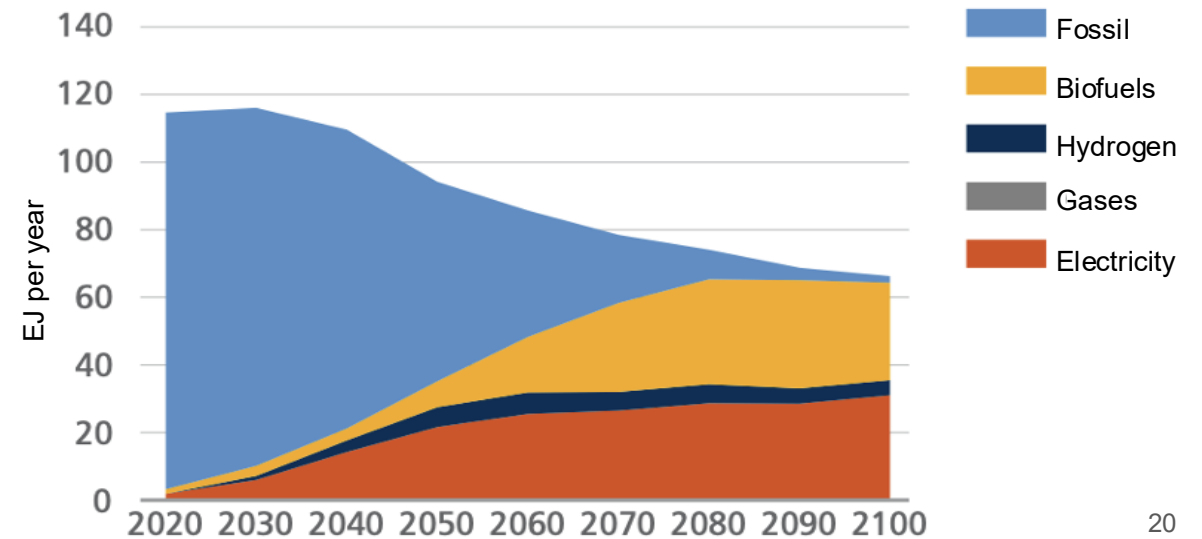
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

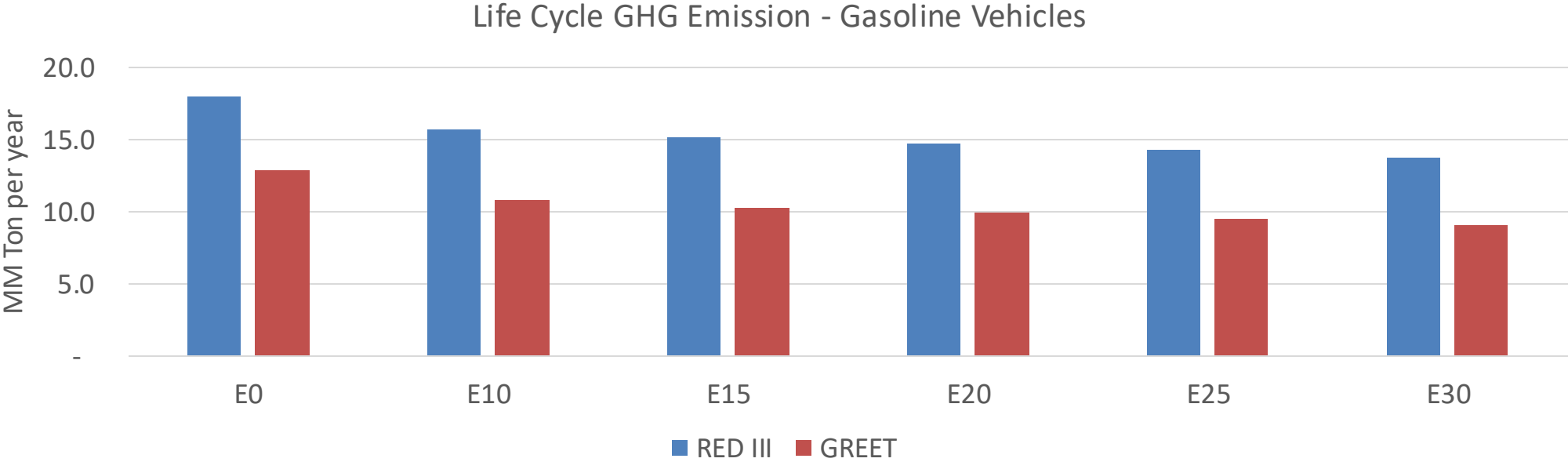
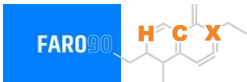
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Dominican Republic – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	51.6
2035 Target (MMT/year)	36.1
Delta	15.5

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	16.4%	19.9%	22.9%	25.8%	29.4%
RED II/III	0.0%	12.7%	15.6%	18.1%	20.6%	23.5%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	13.7%	16.6%	19.1%	21.5%	24.5%
RED II/III	0.0%	14.8%	18.2%	21.1%	23.9%	27.4%

Source: Greet, RED II/III, climateactiontracker.org, F90